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Scenario reduction and scenario tree generation for stochastic programming using Sinkhorn distance

Sanjula Kammammettu, Zukui Li *

Department of Chemical and Materials Engineering, University of Alberta, 9211 116 St, Edmonton, Alberta, Canada T6G1H9

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ABSTRACT

Scenario-based stochastic programming is a widely used method for optimization under uncertainty. The solution quality of this approach is dependent on the approximation of the underlying uncertainty distribution. Therefore, the optimal generation of scenarios (or scenario trees) is a pertinent research objective in stochastic programming. In this work, we approach the scenario reduction and scenario tree generation problem through the perspective of optimal transport, specifically entropy-regularized optimal transport. The availability of an iterative procedure to compute the optimal entropy-regularized transport plan between support sets, using the Sinkhorn–Knopp algorithm in lieu of conventional linear programming-based optimal transport, is found to decrease solution time appreciably, with a decrease in memory burden as well. We present algorithms for optimal scenario reduction and multistage scenario tree generation, and illustrate their use through two case studies. We show that the proposed approach generates high-quality scenarios whose use in stochastic programming offers solutions with good accuracy.

1. Introduction

Industrial processes generally operate under a varying range of uncertainties, from demand and supply, to resource availability and scheduling. Optimal decision-making, therefore, must take into account the underlying uncertainty in order to obtain practically feasible decisions. In this regard, it is imperative to assess the amount of distributional information about the uncertain parameters, if any, available for incorporation into optimization.

Two popular optimization approaches under uncertainty are robust optimization and stochastic programming. Robust optimization uses the support information about the underlying uncertainty in the optimization problem, and solves the problem for the worst case realization of uncertainty, thus offering a risk-averse solution. While this approach is often used in applications concerning process safety and design, robust optimization also leads to markedly higher costs. In contrast, stochastic programming, proposed by Dantzig (1955), comprises methods in which the probability distribution of the uncertain parameter is assumed to be known, or reasonably estimated from available process history. In order to solve the stochastic programming problem numerically, a deterministic equivalent of the problem may be obtained through scenario-based approximation method. In this method, the optimization problem is solved for a finite set of scenarios generated to represent the uncertainty in discrete realizations with corresponding probabilities of occurrence. Generally, a large number of scenarios would capture the true uncertainty distribution better than a smaller number. However, it is found that as the number of scenarios increases, the problem complexity, as well as solution time, increases greatly. Therefore, there is a need to explore methods of optimal scenario reduction.

The concept of scenario reduction was pioneered by Dupačová et al. (2003), who proposed the use of a natural probability metric as an approximation metric in order to obtain the closest subset from a larger superset of scenarios. The authors proposed the use of Fortet-Mourier metrics for the reduction of electrical load scenario trees used for power management under uncertainty. The stability of multistage stochastic programs was analyzed by Heitsch et al. (2006) who posited that the reduction of multistage scenario trees should be based on L_r distances as well as filtration distances. Subsequently, Heitsch and Römisch (2009) derived algorithms for the reduction of multistage scenario trees using this idea. Xu et al. (2012) developed algorithms using K-means clustering and LP moment-matching methods to approximate multistage scenario trees from a large scenario fan description of the uncertain parameter. In this method, the authors generated a multistage scenario tree from a scenario fan composed of a large number of profiles evolving in time, generated as a result of random processes. At each stage, the superset of points is reduced to a smaller subset comprising cluster centers chosen by the K-means clustering technique. Chen

* Corresponding author.
E-mail address: zukui@ualberta.ca (Z. Li).

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and Yan (2018) designed a scenario tree reduction algorithm through clustering tree nodes based on a new distance function to measure the difference between two scenario trees.

Li and Floudas (2014) treated the scenario reduction problem as a mixed integer linear programming (MILP) problem. The authors designed an MILP-based method for scenario reduction using Kantorovich Distance. In this method, the binary variables denote whether the scenario is retained or removed to meet the new reduced scenario set size. The constraints in the MILP model pertain to the conservation of probability mass from the superset to the reduced subset. This is a single-step approach in the sense that this optimization problem is solved once, and the optimal reduced scenario subset is obtained. In contrast, Li and Li (2016) proposed a computationally efficient iterative algorithm for scenario reduction that circumvents the limitations posed by the aforementioned MILP formulation through the use of transportation distance. In a recent work by Zhou et al. (2019), the authors introduce an improved method using K-means with a typicality degree approach for scenario reduction.

In addition to transportation metric-based techniques, scenario reduction has also been accomplished for decision-making under uncertainty using other methods. Karuppiah et al. (2010) proposed a heuristic scenario selection strategy that the overall probability of occurrence of a particular realization of any uncertain parameter in the final set of scenarios should be equal to the probability of the uncertain parameter taking on that particular value. Meira et al. (2016) performed scenario reduction using representative models in oil fields. Arpón et al. (2018) have chosen the Conditional-Value-at-Risk (CVaR) measure as the objective function in order to accomplish the reduction. In the work of Hu and Li (2019), they balanced the objective of scenario reduction by maximizing the likeness between the original scenario superset and the reduced subset, while simultaneously minimizing the correlation loss before and after the reduction process. Silvente et al. (2019) proposed a scenario tree reduction method using sensitivity analysis for nonlinear optimization models. In their work, the authors chose the sensitivity of scenarios as a measure of identifying which scenarios to retain in a larger superset. Scenario reduction has also been approached through the purview of machine learning. Li and Gao (2019) used deep learning to reduce scenario supersets. In their work, the authors transformed scenarios into a format similar to that of images, and fed these transformations to a deep convolutional neural network to obtain a similar image-like output of the reduced subset. Medina-Gonzalez et al. (2020) proposed a scenario-reduction method that integrates data mining, graph theory and community detection concepts to represent the uncertain information as a network and identify the most efficient communities/clusters. Bounitsis et al. (2022) proposed a data-driven MILP model for the distribution matching problem behind scenario reduction. They have shown improved efficiency through integration of copula-based simulation and clustering method.

In this work, we aim to improve the LP-based scenario reduction strategy, proposed by Li and Li (2016) by replacing the inner linear optimal transport problem with entropy-regularized optimal transport solved using the Sinkhorn–Knopp algorithm (Section 3). We present the algorithm for optimal scenario reduction using Sinkhorn distance (Section 4), and extend its application to the optimal generation of multistage scenario trees for stochastic programming (Section 5). Furthermore, we use the process distance metric (Pflug and Pichler, 2012) to evaluate the approximation quality between the generated scenario tree, and the original scenario fan. Finally, we illustrate the use of these algorithms on two case studies (Section 6).

2. Optimal transport distance between distributions

The extent of similarity between two probability distributions is quantified in literature through a number of measures, such as Kullback–Leibler Divergence (KLD), Hellinger Distance, power distance, and correlation similarities. However, there are certain instances where measures like KLD cannot be used, as is the case when the level of similarity between two distributions defined on different probability spaces is to be quantified.

When distributions defined on different support sets are to be compared, the problem can be viewed through the lens of optimal transport. Optimal transport was first studied by Monge (1781) in the context of transportation of mined soil from the quarry to various construction sites, famously titled the “earth-mover’s problem” in literature. The optimal transport problem aims to find the most efficient manner in which to transfer probability mass from the elements of one distribution to another, while minimizing a chosen cost function. It follows intuitively that if the two distributions in question are vastly dissimilar, the task of moving the probability masses from one distribution to the other requires more effort, which is quantified by the optimal transport distance.

In the pioneering work of Monge (1781), a continuous transport map $f(x)$ was estimated between two probability distributions $P(x)$ and $Q(y)$, with support sets X and Y, respectively, so as to minimize the expected cost of transportation $c(x, f(x))$, as given by Eq. (1),

$$M(P,Q) = \inf_{f \in MP} \int_X c(x, f(x)) dP(x) \tag{1}$$

MP denotes the set of all mappings that preserve the transfer of probability from $P(x)$ to $Q(y)$ for any Borel subset A of Y (Eq. (2)).

$$MP = \left\{ f : X \to Y \Big| \int_{f^{-1}(A)} dP(x) = \int_A dQ(y) \right\} \tag{2}$$

It must be noted that the Monge formulation of the optimal transport problem does not allow for probability masses to be split during transfer.

2.1. Kantorovich distance and Wasserstein distance

Kantorovich (1942) approached the optimal transport problem through a relaxed formulation, allowing for the splitting for probability masses; this formulation uses the concept of a transport plan which is given by (Eq. (3)),

$$K(P,Q) = \inf_{\pi \in \Pi(P,Q)} \int_{X \times Y} c(x,y) d\pi(x,y) \tag{3}$$

Here, $\Pi(P,Q)$ is the set of all joint distributions whose marginal distributions are $P(x)$ and $Q(y)$.

The optimal transport problem can be formulated as a linear programming (LP) problem for transport between discrete distributions. Consider a simplified version of the same (Model (4)),

$$\min_{\pi_{i,j}} \quad \sum_{i,j} c_{i,j} \pi_{i,j} \tag{4a}$$

$$\text{s.t.} \quad \sum_j \pi_{i,j} = a_i \quad \forall i \tag{4b}$$

$$\sum_i \pi_{i,j} = b_j \quad \forall j \tag{4c}$$

$$\pi_{i,j} \geq 0 \quad \forall i,j \tag{4d}$$

In Model (4), the variables $\pi_{i,j}$ are elements of the optimal transport plan, and represent the amount of probability mass that is transferred from one distribution’s elements ($x_i \sim P(x)$) to those of another distribution’s ($y_j \sim Q(y)$). The cost of moving probability mass between elements x_i and y_j is computed as $c_{i,j}$, which is often computed using the norm $c(x,y) = \|x - y\|$. For example, 1-norm and 2-norm based distance for two n-dimensional points x and y is given as: $\|x - y\|_1 = \sum_{k=1}^{n} |x_k - y_k|$, and $\|x - y\|_2 = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2}$.

The objective function in (4a) represents the total cost in moving probability masses from $P(x)$ to $Q(y)$ to be minimized for optimal transport. The constraints in Model (4) represent the conservation of probability mass with respect to $P(x)$ and $Q(y)$, respectively, subject to non-negativity constraints on $\pi_{i,j}$. The vectors a and b in Eqs. (4b) and (4c) contain the probability masses of the elements $x_i \sim P(x)$ and $y_j \sim Q(y)$, respectively. It must be noted that the transport plan obtained through the Kantorovich formulation of the optimal transport problem is not a one-to-one mapping, ergo allowing for the probability mass of one discrete element in the source distribution to be split across that of multiple elements in the destination.

The above optimal transport problem formulation is a special case of the p-Wasserstein distance ($p \geq 1$), which is the optimal objective of the following optimal transport problem (Eq. (5)),

$$W_p(P,Q) = \left(\inf_{\pi \in \Pi(P,Q)} \int_{X\times Y} c(x,y)^p d\pi(x,y) \right)^{\frac{1}{p}} \tag{5}$$

For $p = 1$, the 1-Wasserstein distance is termed as the Kantorovich Distance (KD), which is the optimal objective of the optimal transport problem in Model (4).

3. Entropic regularization and Sinkhorn distance

The optimal transport plan $\pi_{i,j}$ in Model (4) is an $M \times N$-dimensional decision variable, where M and N are the sizes of the support sets X and Y. For large dimensional optimal transport problems, the formulation in Model (4) encounters a memory bottleneck for computation of the optimal transport plan . To mitigate this issue, Lellmann et al. (2014) used the concept of Kantorovich–Rubinstein duality in which the dual formulation of the optimal transport problem is solved. Another method to tackle the memory burden in conventional optimal transport makes use of entropy regularization of the problem (Cuturi, 2013) that be solved using an iterative approach, rather than a linear programming-based formulation.

A matrix with lower entropy has most of its non-zero values concentrated in fewer points, while a matrix with higher entropy is smoother and has a more uniform distribution of its non-zero values across its elements. Therefore, entropy regularization is carried out to make the coupling matrix smoother by introducing the regularization coefficient (γ) and the entropy of the transport plan ($H = -\sum_{i,j} \pi_{i,j}[\log(\pi_{i,j})-1]$) to the objective function (Peyré et al., 2019). Notice that the objective is of minimization type and we use negative entropy such that the entropy can be maximized to achieve smooth transport plan. The entropy-regularized optimal mass transport problem is given as Model (6),

$$\mathrm{SD} = \min_{\pi_{i,j}} \quad \sum_{i,j} c_{i,j}\pi_{i,j} + \gamma \pi_{i,j}[\log(\pi_{i,j}) - 1] \tag{6a}$$

$$\text{s.t.} \quad \sum_{j} \pi_{i,j} = a_i \qquad \forall i \tag{6b}$$

$$\sum_{i} \pi_{i,j} = b_j \qquad \forall j \tag{6c}$$

$$\pi_{i,j} \geq 0 \qquad \forall i,j \tag{6d}$$

The optimal objective value of the entropy-regularized optimal transport problem in Model (6) is referred to as the Sinkhorn Distance (SD). Note that SD is not a metric since it is biased: $SD(P,P) \neq 0$.

The smoothness of the coupling matrix increases with the value of γ. Therefore, lower the value of γ, closer the solution to that of the original optimal transport problem (Peyré et al., 2019). The entropy-regularized transport problem can be solved using the Sinkhorn–Knopp algorithm (Peyré et al., 2019; Sinkhorn, 1964).

The Lagrangian for Model (6) is given as (Eq. (7)),

$$L = \left\{ \sum_{i,j} c_{i,j}\pi_{i,j} + \gamma \sum_{i,j} \pi_{i,j}[\log(\pi_{i,j}) - 1] \right\} + \boldsymbol{\alpha}^\top [\boldsymbol{\Pi}(\mathbf{1}_{N\times 1}) - a] + \boldsymbol{\beta}^\top [\boldsymbol{\Pi}^\top(\mathbf{1}_{M\times 1}) - b] \tag{7}$$

where $\boldsymbol{\alpha}_{M\times 1}$ and $\boldsymbol{\beta}_{N\times 1}$ are Lagrange multiplier vectors corresponding to constraints (6b) and (6c), respectively, and $\boldsymbol{\Pi}$ is the matrix $[\pi_{i,j}]$. Based on stationary conditions, differentiating the Lagrangian with respect to the transport plan, we have (Eq. (8)),

$$\frac{\partial L}{\partial \pi_{i,j}} = c_{i,j} + \gamma \log(\pi_{i,j}) + \alpha_i + \beta_j = 0 \tag{8}$$

Solving for $\pi_{i,j}$, the expression for the optimal transport plan is obtained as (Eq. (9)),

$$\underbrace{\pi_{i,j}}_{\boldsymbol{\Pi}} = \underbrace{\exp\left(-\frac{\alpha_i}{\gamma}\right)}_{\boldsymbol{U}} \underbrace{\exp\left(-\frac{c_{i,j}}{\gamma}\right)}_{\boldsymbol{M}} \underbrace{\exp\left(-\frac{\beta_j}{\gamma}\right)}_{\boldsymbol{V}} \tag{9}$$

When the entries of the optimal transport plan are viewed in matrix form, Eq. (9) shows that the optimal transport plan $\boldsymbol{\Pi}$ is obtained from a positive kernel cost matrix $\boldsymbol{M}$ scaled by diagonal matrices $\boldsymbol{U}$ and $\boldsymbol{V}$. For a matrix $\boldsymbol{M}$ with positive entries, there exist two diagonal matrices, $\boldsymbol{U}$ and $\boldsymbol{V}$, with positive diagonal entries such that the matrix product $\boldsymbol{UMV}$ has the ith row sum a_i and jth column sum b_j - this is termed as the matrix scaling problem (Nemirovski and Rothblum, 1999). The corresponding iterative algorithm for computing vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ (constructed using diagonal entries of $\boldsymbol{U}$ and $\boldsymbol{V}$, respectively) is given as (Eq. (10)),

$$\boldsymbol{u}^{(g+1)} = \frac{\boldsymbol{a}}{\boldsymbol{M}\boldsymbol{v}^{(g)}} \qquad \boldsymbol{v}^{(g+1)} = \frac{\boldsymbol{b}}{\boldsymbol{M}^\top \boldsymbol{u}^{(g+1)}} \tag{10}$$

Here, g represents the iterations in the calculation of $\boldsymbol{u}$ and $\boldsymbol{v}$. The proof of convergence of this iterative procedure (Deming and Stephan, 1940; Bacharach, 1965) was worked upon by Sinkhorn (1964). The errors associated with the vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ are given by (Eq. (11)),

$$\epsilon_a = \frac{\|\boldsymbol{u}\boldsymbol{M}\boldsymbol{v} - \boldsymbol{a}\|}{\|\boldsymbol{a}\|} \qquad \epsilon_b = \frac{\|\boldsymbol{v}\boldsymbol{M}^\top\boldsymbol{u} - \boldsymbol{b}\|}{\|\boldsymbol{b}\|} \tag{11}$$

The vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ are iteratively computed until both error terms ϵ_a and ϵ_b are below the specified error threshold, $\bar{\epsilon}$. The entries of the optimal transport plan $\pi_{i,j} \in \Pi$ are calculated using Eq. (9), where the final $\boldsymbol{u}$ and $\boldsymbol{v}$ vectors comprise the principal diagonal elements of matrices $\boldsymbol{U}$ and $\boldsymbol{V}$, respectively. Finally, the Sinkhorn Distance (SD) is computed as (Eq. (12)),

$$\mathrm{SD} = \sum_{i,j} c_{i,j}\pi_{i,j} + \gamma \sum_{i,j} \pi_{i,j}[\log(\pi_{i,j}) - 1] \tag{12}$$

A step-wise procedure for the calculation of Sinkhorn distance between two sets I and S is outlined in Algorithm 1.

Next, we illustrate an instance of entropy-regularized (SD-based) optimal transport. Here, the optimal transport problem entails the movement of probability point masses of one distribution, represented by the blue dots in the outer annulus, to those of another distribution, represented by the red dots in the inner spiral, in Fig. 1, using the Sinkhorn distance iterative calculations detailed in Algorithm 1.

Fig. 1 illustrates the effect of the regularization coefficient γ on the smoothness of the optimal transport plan. The solid lines represent maximum probability transfers, while the dotted lines represent probability mass transfer greater than 5%. Here, the optimal transport problem was solved for four values of $\gamma = 10, 1, 0.1$ and 0.01. It is observed that as the value of γ decreases, the smoothness of the optimal transport plan reported decreases as well. This is illustrated by the gradual disappearance of the dotted lines as γ increases, thereby showing that the optimal transport plan at lower values of γ mirrors the plan obtained from conventional optimal transport ($\gamma = 0$), and contains a number of zero values.

Choice of regularization coefficient (γ): In the entropy-regularized optimal transport problem, γ is introduced to smoothen the optimal transport plan such that the number of non-zero elements reduces with an increase in γ. In practice, γ is chosen to be sufficiently small so as to best approximate the conventional optimal transport plan, while large enough to avoid numerical issues in the computation of the exponential terms involved in the Sinkhorn algorithm (Algorithm 1).

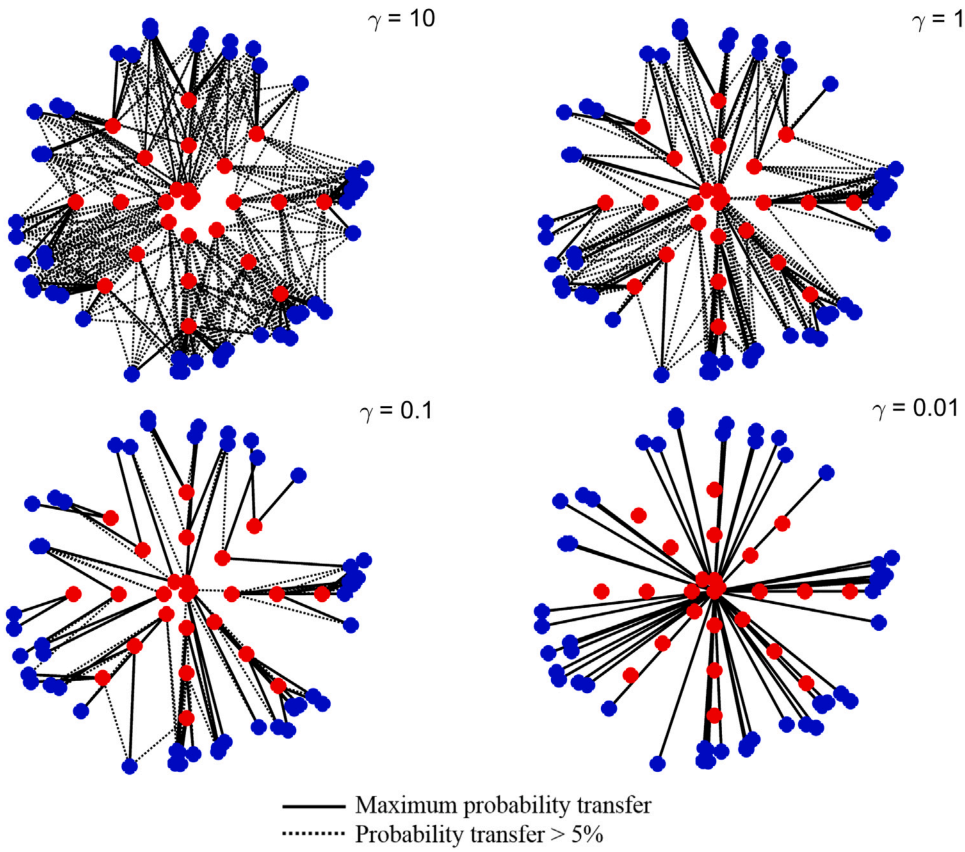

Fig. 1. Effect of the regularization coefficient (γ) on smoothness of the transport plan.

Algorithm 1: Sinkhorn Algorithm for Optimal Transport with Entropic Regularization

Input: Set $I = \{x_i\}$, with probability a_i for element $x_i \in X, x_i \sim P(x)$

Set $S = \{y_j\}$, with probability b_j for element $y_j \in Y, y_j \sim Q(y)$

Error threshold $\bar{\epsilon}$

Procedure:

Step 1: Evaluate cost matrix entries $c_{i,j} := ||x_i - y_j||_2$

Step 2: Evaluate kernel function entries $m_{i,j} := \exp(-\frac{c_{i,j}}{\gamma})$

Step 3: Initialize iteration: $g \leftarrow 1$

Initialize vector $\boldsymbol{v}^{(g)}$ (value 1 for all entries)

while $\epsilon_a, \epsilon_b \geq \bar{\epsilon}$ **do**

Evaluate vectors $\boldsymbol{u}^{(g+1)} = \frac{\boldsymbol{a}}{\boldsymbol{M}\boldsymbol{v}^{(g)}}, \boldsymbol{v}^{(g+1)} = \frac{\boldsymbol{b}}{\boldsymbol{M}^\top \boldsymbol{u}^{(g+1)}}$

Evaluate errors $\epsilon_a = \frac{||\boldsymbol{u}^{(g+1)}\boldsymbol{M}\boldsymbol{v}^{(g+1)} - \boldsymbol{a}||}{||\boldsymbol{a}||}$, $\epsilon_b = \frac{||\boldsymbol{v}^{(g+1)}\boldsymbol{M}^\top \boldsymbol{u}^{(g+1)} - \boldsymbol{b}||}{||\boldsymbol{b}||}$

$g \leftarrow g + 1$

end

Step 4: Calculate optimal transport plan $\boldsymbol{\Pi} = [\pi_{i,j}] = \boldsymbol{UMV}$

Step 5: Calculate Sinkhorn distance

$$\mathrm{SD} = \textstyle\sum_{i,j} c_{i,j}\pi_{i,j} + \gamma \sum_{i,j} \pi_{i,j}[\log(\pi_{i,j}) - 1]$$

Output: Sinkhorn distance SD

Optimal transport plan $\pi_{i,j}$

4. Scenario reduction using Sinkhorn distance

In this section, we propose an algorithm for optimal scenario reduction using Sinkhorn distance for a large superset of points, denoted by I, to obtain a reduced subset S using an iterative method where the extent of similarity between I and S is computed using the Sinkhorn distance. Section 4.1 presents the problem through a mixed integer nonlinear optimization (MINLP) formulation. Section 4.2 presents an iterative solution algorithm for this problem using the Sinkhorn distance calculation detailed in Algorithm 1, followed by a numerical illustration of the algorithm.

4.1. MINLP model

The mixed integer linear programming (MILP) model presented in Li and Floudas (2014) may be used to reduce a superset of scenarios to an optimal reduced subset. This optimization model minimizes the Kantorovich Distance (KD) between the superset and subset. In a similar manner, a mixed integer nonlinear programming (MINLP) formulation of the entropy regularized optimal transport problem may be developed, as shown in Model (13).

$$\min_{\pi_{i,i'}, b_{i'}, y_i} \quad \sum_{i \in I} \sum_{i' \in I} \pi_{i,i'} c_{i,i'} + \sum_{i \in I} \sum_{i' \in I} \gamma \pi_{i,i'} [\log(\pi_{i,i'}) - 1] \tag{13a}$$

$$\text{s.t.} \quad \sum_{i'} \pi_{i,i'} = a_i \qquad \forall i \in I \tag{13b}$$

$$\sum_{i \in I} \pi_{i,i'} = b_{i'} \qquad \forall i' \in I \tag{13c}$$

$$\sum_{i' \in I} b_{i'} = 1 \tag{13d}$$

$$\sum_{i \in I} y_i = r \tag{13e}$$

$$\epsilon y_i \leq b_i \leq y_i \qquad \forall i \in I \qquad (13f)$$

$$\pi_{i,i'} \geq 0 \qquad \forall i, i' \in I \qquad (13g)$$

$$b_{i'} \geq 0 \qquad \forall i' \in I \qquad (13h)$$

$$y_i \in \{0,1\} \qquad \forall i \in I \qquad (13i)$$

In this model, r is the desired number of scenarios in the reduced set, the binary variable y_i denotes the status of retention of the scenario $i \in I$ in the optimal subset. The variable b_i denotes the amount of probability mass assigned to the retained scenario i. Finally, $\pi_{i,i'}$ denotes the amount of probability mass transported between the elements $i, i' \in I$. The parameter ϵ is taken to be a small value of the order of magnitude 10^{-9}, in order to define the relationship between the binary status variable y_i and assigned probability mass b_i.

In addition to utilizing a nonlinear objective function to be minimized, Model (13) contains binary variables whose number increases as the size of the superset to be reduced increases, thus escalating the problem complexity and computational time involved. This issue of rapidly surging problem complexity with the number of integer variables presents a clear need for an iterative solution of a simplified optimization problem. It may also be noted that an MILP formulation for scenario reduction based on KD can be easily obtained after removing the second term in the objective function.

4.2. Solution algorithm

The Sinkhorn distance calculation presented in Algorithm 1 details the procedure for transporting probability masses of the elements of one distribution (with finite support set I) to the other distribution (with support set S). Therefore, SD may be viewed as a quantitative measure of dissimilarity between the two distributions: a smaller value of SD is indicative of more similar probability distributions. This interpretation of the optimal transport problem can be extended to finding the optimal S^* such that SD(I, S) is as small as possible. To this end, we propose the following iterative procedure for optimal scenario reduction in Algorithm 2.

The scenario superset I is fed as input to this algorithm, and it is to be reduced to an optimal subset S^* containing r scenarios. At every iteration (d), the Sinkhorn distance between I and $S^{(d)}$ is computed using Algorithm 1. Using the maximum probability transfer links from the optimal transport plan $\Pi^{(d)}$, clusters are formed around the points retained in $S^{(d)}$. The relative error between the Sinkhorn distance values at successive iterations is computed; this value is compared with the specified threshold value (β_{SD}) as the termination criterion. If the computed relative error is above the threshold, the cluster centers are updated to give minimum in-cluster transportation distance, and this updated subset is used in the following iteration. The optimal subset S^* is obtained from the final iteration of the algorithm. In this work, we assume that all points in the superset have an equal probability of occurrence. The probabilities of the optimally retained scenarios in S^* are computed using the cardinality of the optimal clusters. The optimal subset S^* with the corresponding probability vector, and the final clusters $C_1, C_2, \ldots, C_r$, are reported as outputs. Note that each cluster contains the values as well as the indices of the points with respect to the superset I.

In the scenario reduction problem addressed in this work, the number of scenarios (r) in the optimally reduced subset is a fixed, user-defined parameter. Ideally, through the law of large numbers, a large r ensures that the optimal solution to the scenario-based problem tends to the true optimal solution. However, too large a value of r might make the problem computationally difficult to solve. There are a few results in literature which provide the sample size determination method. For example, Kleywegt et al. (2002) provided a bound on the sample size required to find an ϵ-optimal solution with probability at least $1 - \alpha$.

Algorithm 2: Optimal scenario reduction using Sinkhorn distance

Input: Superset I, Subset size r, Error threshold β_{SD}
Procedure: Initialize iteration $(d \leftarrow 1)$
while *RelativeError* $\geq \beta_{SD}$ **do**
 Step 1: Define subset (S)
 if $d = 1$ **then**
  $S^{(d)}$ is set as r randomly chosen points in I
 else
  $S^{(d)} := S^{(d-1)}$
 end
 Step 2: Calculate Sinkhorn distance (SD$^{(d)}$) and optimal transport plan $(\Pi^{(d)})$ using *Algorithm 1*
 Step 3: Form clusters $C_1, C_2, \ldots, C_r$ around points in S, using maximum probability transfer links in $\Pi^{(d)}$
 Step 4: Calculate relative error
 if $d = 1$ **then**
  Proceed to step 5
 else
  $$\text{RelativeError} = \frac{\text{SD}^{(d-1)} - \text{SD}^{(d)}}{\text{SD}^{(d-1)}}$$
 end
 Step 5: Update cluster centers $S^{(d)}$ as points with minimum in-cluster transportation distance
 $d \leftarrow d + 1$
end
Step 6: Obtain final reduced subset $S^* := S^{(d)}$, and corresponding clusters $C_1, C_2, \ldots C_r$
Step 7: Compute probabilities of points in optimal reduced subset S^* as

$$p_r = \frac{\text{Number of points in cluster } C_r}{\text{Number of points in superset } I}$$

Output: Final reduced subset S^* containing r points, the corresponding probability and clusters $C_1, C_2, \ldots, C_r$

Illustrating example

In this section, we present a numerical illustrative example of the proposed optimal scenario reduction algorithm using Sinkhorn distance (Algorithm 2) presented in Section 4.2. Here, a superset I comprising 20 points (blue) is to be reduced to a subset containing 3 points. Initially, a random subset S containing 3 points (red) is chosen (Fig. 2a). The Sinkhorn distance between I and S is calculated, using Algorithm 1, as 1.3379, and links (black) are established between points in the superset and subset, representing the clusters formed using the optimal transport plan. The subset S is updated (S^{new}) as the points in each cluster that give the minimum in-cluster transportation distance. The SD between I and S^{new} is calculated to be 0.9824 (Fig. 2b). The relative error in Sinkhorn distance is found to be 26.57%. In this example, the chosen threshold (β_{SD}) for relative error is 5%, that is, the clusters are successively updated until the relative error in Sinkhorn distance is below 5%. In the next iteration, the SD between I and the updated S^{new} is 0.9184, and the new relative error in SD is 6.52%. In the following iteration, it is observed that the updated subset does not change from that of the previous iteration; here, the SD remains 0.9184 and the relative error in SD is, thus, 0%. Therefore, the subset from the final iteration is chosen as the optimal subset of scenarios (Fig. 2c).

4.3. Computational study

Fig. 3 depicts the evolution in computational time for scenario reduction of increasingly large superset sizes, from 1000 to 20 000 scenarios, to an optimal subset containing 50 scenarios. The figure compares the results from the proposed SD-based reduction in this work, to the LP-based scenario reduction proposed by Li and Li (2016).

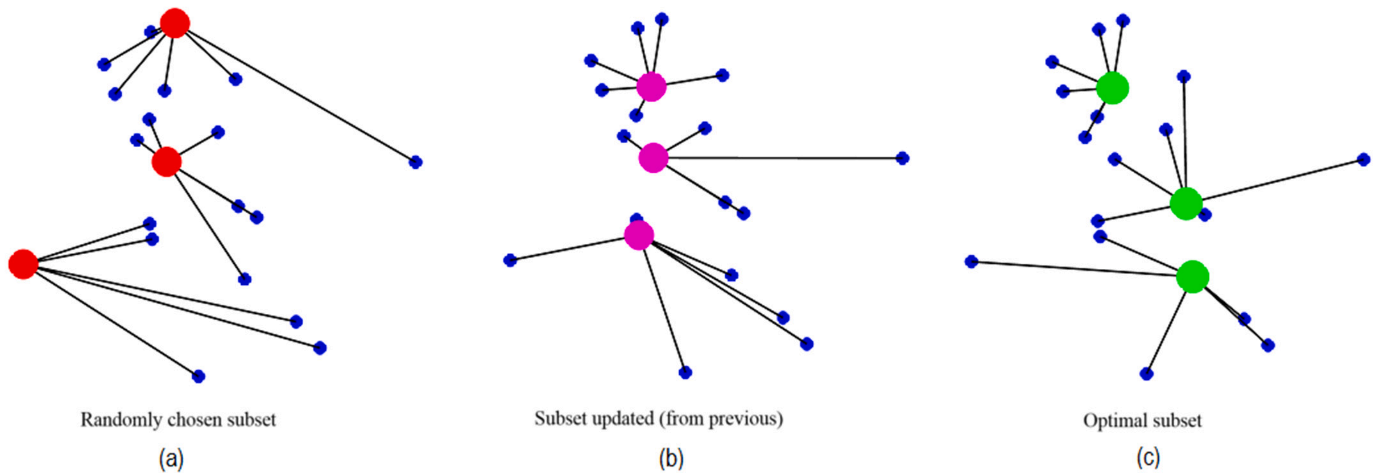

Fig. 2. Illustration: optimal scenario reduction using Sinkhorn distance.

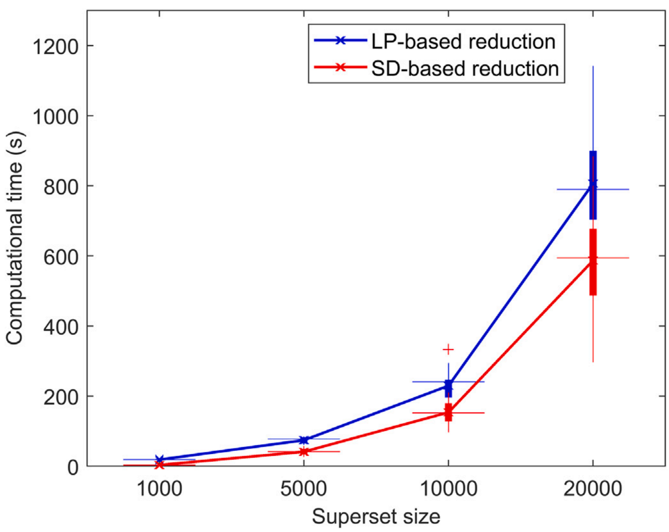

Fig. 3. A comparison of computational times using the LP-based algorithm (Li and Li, 2016), and the proposed SD-based approach.

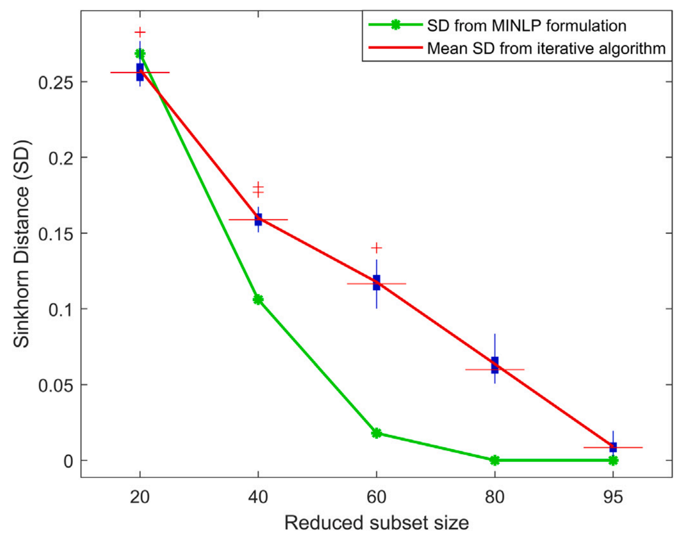

Fig. 4. A comparison of Sinkhorn distance between a 100 scenario superset and reduced size subsets via the MINLP approach and the proposed SD-based algorithm for scenario reduction.

The LP-based reduction is found to be computationally slower than the proposed SD-based approach. Both scenario reduction schemes were tested on a desktop computer running on Intel(R) Core(TM) i5 CPU @3.2 GHz, 4 cores with 8 GB of physical RAM.

Fig. 4 compares the Sinkhorn distance between a superset of 100 normally distributed scenarios to a number of reduced size subsets obtained through the MINLP approach in Model (13), and the proposed iterative SD-based algorithm (Algorithm 2). Since the iterative approach utilizes a random starting subset to begin the scenario reduction process, 25 trials were conducted to obtain an average SD for this approach, as depicted by the boxplot. The numerical results for Fig. 4 are presented in Table 1. The MINLP formulation runs to the default GAMS resource limit of 1000 s, while each trial of the proposed iterative SD-based algorithm takes less than 10 s. It must further be noted that the MINLP formulation was run on GAMS using the BONMIN (Bonami and Lee, 2011) solver, which reports the locally optimal solution.

Table 2 presents a summary of different scenario reduction approaches through the lens of optimal transport. The entropy-regularized formulation of the optimal transport problem enjoys a computational time advantage due to the availability of the Sinkhorn–Knopp iterative algorithm to obtain a smoother optimal transport plan, as compared to the conventional formulation, especially when the datasets involved are high-dimensional in nature. The exact scenario reduction problem utilizes binary variables that depict the status of retention or removal of scenarios from the superset; in either the conventional, or the entropy-regularized formulations, the optimization problem suffers from a slow solution process. To this end, an iterative LP-based algorithm was developed by Li and Li (2016). The iterative Sinkhorn distance-based approach to scenario reduction presented in this section is shown to perform faster than the iterative LP-based method (Li and Li, 2016), as shown in Fig. 3.

Table 1
A comparison of evolution of Sinkhorn distance for the MINLP approach versus the proposed iterative SD-based algorithm for scenario reduction.

Size of superset	Size of reduced subset	SD (MINLP approach)	Mean SD (proposed iterative algorithm)
	20	0.2686	0.2580
	40	0.1061	0.1601
100	60	0.0180	0.1178
	80	0	0.0636
	95	0	0.0091

Table 2
A summary of scenario reduction approaches through optimal transport.

	Kantorovich Distance (KD) *Conventional optimal transport*	Sinkhorn Distance (SD) *Entropy-regularized optimal transport*
Distance evaluation problem	LP problem [Model (4)]	NLP problem with convex objective and linear constraints [Model (6)] Efficiently solved using numerical iterations
Exact scenario reduction problem	MILP problem Li and Floudas (2014)	MINLP problem with convex objective and mixed integer linear constraints [Model (13)]
Iterative scenario reduction algorithm	– Integer variables are removed – Each iteration solves an LP formulation of the conventional optimal transport Li and Li (2016)	– Integer variables are removed – Each iteration solves a numerical computation of the entropy-regularized optimal transport problem [Algorithm 2]

5. Multistage scenario tree generation algorithm

In this section, we extend the scenario reduction algorithm presented in Section 4 to the problem of optimal scenario tree generation from a scenario fan. A scenario fan comprises a number of profiles generated as a result of stochastic processes and the task is to obtain a multistage scenario tree to be used for stochastic programming. The algorithm presented in this section uses the procedure outlined in Algorithm 2 in a stage-wise manner to obtain the nodes of the scenario tree. Section 5.1 describes the proposed algorithm for optimal scenario tree generation followed by a numerical illustration of the algorithm, while Section 5.2 introduces the Process Distance (*PD*) measure that is used to evaluate the approximation quality of scenario trees generated using the proposed algorithm.

5.1. Proposed workflow

The generation of a multistage scenario tree from a scenario fan is treated as a sequential extension of the scenario reduction problem, as described in Section 4. Here, a scenario fan is considered to be a collection of profiles generated via stochastic processes over time, originating from a single root. An important convention to be clarified in this work is that of the "stage".

In this work, we refer to the root node of the scenario tree as "stage 0". The subsequent stages are numbered 1, 2, …, and so on. With this notation, a two-stage stochastic programming problem is solved using a scenario tree with stage 0 and stage 1. Similarly, a n-stage stochastic programming problem is solved using a scenario tree with 0, 1, …, $n-1$.

The proposed algorithm 3 for optimal scenario tree generation contains two main loops:

- Time stages ($t = 1, 2, \dots, T$),
- Nodes over each stage ($w = 1, 2, \dots, n_{t-1}$)

Let $B_t, t = 1, 2, \dots, T$ describe the number of leaf nodes that originate out of nodes at stage t. The total number of leaf nodes n_t at each stage t is $n_t = n_{t-1} B_t$. Here, $n_0 = 1$ by default. At each stage t, an optimal set of nodes approximating the superset of nodes at the time step pertaining to that stage in the scenario fan is found. However, the number of such supersets to reduce at each stage depends on the number of nodes present in the previous stages. At each stage $t \geq 1$, there exist multiple supersets to be reduced, denoted by I_w^t, where $w = 1, 2, \dots, n_{t-1}$, each superset corresponding sequentially to the leaf nodes of the previous stage. The use of clusters to generate subsequent supersets, ensures that there is a stage-wise link between the nodes of the generated multistage scenario tree. Every parent node at stage t in the scenario tree branches out to leaf nodes at stage $t+1$; these leaf nodes are obtained through the reduction of supersets containing points from the same profiles for which scenario reduction was previously performed to give the parent node at stage t.

Consider a simple 3 stage example as shown in Fig. 5. The branching structure in this tree is $\{2, 3, 2\}$ for stages 0, 1 and 2, respectively.

Algorithm 3: Optimal scenario tree generation using Sinkhorn distance

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Input: Scenario fan $X_{q,\tau}$ where q: profile number, τ: time step
 Number of stages $t = 1, 2, ..., T$
 Branching B_t
Procedure:
for $t = 1 : T$ **do**
 Compute $n_t = n_{t-1} B_t$, where $n_0 = 1$
 for $w = 1 : n_{t-1}$ **do**
  if $t = 1$ **then**
   Define $I_w^t := X(\text{all}, t)$
  else
   Define $I_w^t := X(q', t)$ where q': profile numbers of points in C_w^{t-1}
  end
  Using *Algorithm 2*, reduce I_w^t to S_w^t, obtain clusters $C_1^t, C_2^t, ..., C_{n_t}^t$ and the corresponding probabilities
 end
end
Output: Scenario tree with stage-wise leaf nodes S_w^t, and corresponding probabilities $Pr(S_w^t)$

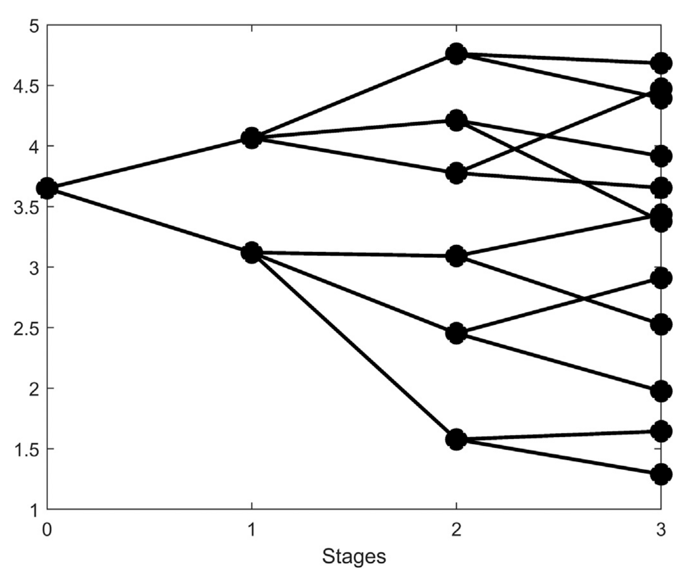


Fig. 5. An example scenario tree.

At stage 1, the superset comprises values from all profiles of the scenario fan at the time step corresponding to stage 1. This superset is to be reduced to $B_1 = 2$ points using Algorithm 2. Therefore, the optimal subset of nodes at stage 1 contains 2 points, and therefore, 2

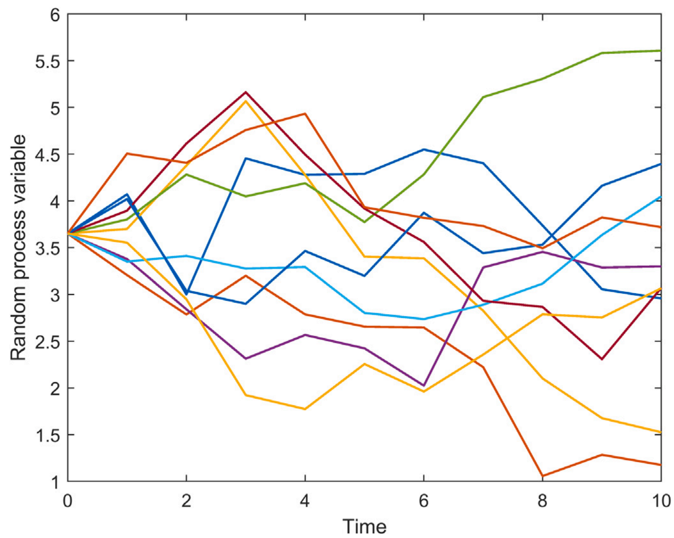


Fig. 6. Scenario fan generation.

clusters, where each point occurs is associated with a probability of occurrence equal to the ratio of number of points in the corresponding cluster and in the superset. At stage 2, two different supersets must be reduced to $B_{t=2} = 3$ points each. These supersets comprise values from those profiles clustered around each node at the previous stage $t = 1$, at the time step corresponding to stage 2. Similarly, at stage 3, six different supersets corresponding to each of the leaf nodes at stage 2, must be reduced to $B_{t=3} = 2$ points each.

Illustrating example

Random parameter occurrences can be described by a scenario fan simulation over time, as shown in Fig. 6. Each strand of the scenario fan depicts a random occurrence of the parameter over the entire time horizon. This scenario fan can be assimilated into a simpler scenario tree structure.

The scenario fan in Fig. 6 contains 10 profiles, and is generated using a base value of 3.65 with a random walk model, as follows,

$$x_{t+1} = x_t + 0.25e \tag{14}$$

Here, e is a random number generated from the standard Gaussian distribution.

In this example, the scenario fan $X_{q,\tau}$ is a matrix containing 10 rows, and 11 columns spanning time steps 0 to 10. The task is to construct a 2-stage scenario tree from this fan, where each stage occurs after 5 time steps; i.e., columns 6 and 11 correspond to stages 1 and 2. It must be noted that the intermediate values or information between stages is not taken into account while generating the scenario tree, and only values at the time steps (τ) corresponding to the stages (t) are used. At each stage, we consider the branching to be $B_t = 2$.

Using the algorithm presented in Section 5.1, the multistage scenario tree (Fig. 7) is obtained as follows: we start with stage $t = 1$. The number of nodes at stage $t = 1$ is given by $n_1 = n_0 B_1$. The number of nodes at stage 0 (n_0) is taken to be 1, by default, and this node occurs with a probability of 1. Therefore, $n_1 = 1 \times 2 = 2$. In order to obtain the nodes at stage 1 of the scenario tree, it is necessary to assess the number of supersets to be reduced at stage 1; at any stage t there are as many supersets to reduce as there are nodes in the previous stage n_{t-1}. Therefore, at stage 1, there is $n_0 = 1$ superset to reduce, and the inner loop variable, w, spans just 1 iteration. Each superset is denoted by I_w^t where w denotes the superset index corresponding to each node at the previous stage, sequentially, and t denotes the stage. At stage 1, since there is only one superset, I_1^1 comprises the values from all rows of X whose column pertains to stage 1, i.e., $X_{[1,2,\dots,10],6}$ (yellow points in Fig. 7(a)). The points in the superset I_1^1 are reduced to a subset containing $B_1 = 2$ points using Algorithm 2, which gives the optimal subset containing the points {2.6560, 4.2895}, with probabilities [0.5,0.5], as well as clusters C_1^1 and C_2^1 formed around the two points (Fig. 7(b)). These clusters are formed based on the maximum probability transfer links in the optimal transport plan. Each cluster contains the profile numbers q' as well the values of the points at t. Here, the point 2.6560 is the cluster center for C_1^1 containing profile numbers (2,4,6,8,10) with corresponding values at $t = 1$ as {2.6560,2.4237,2.8028,3.1973,2.2557}, and the point 4.2895 for C_2^1 containing profile numbers (1,3,5,7,9), with values {4.2895,3.4032,3.7732,3.9175,3.9316}, respectively. The profile numbers in these clusters determine the profiles comprising each superset in the following stage. The corresponding links between parent and leaf nodes at each stage are made, and the probabilities associated with the leaf nodes are recorded from Algorithm 2.

At stage $t = 2$, there are $n_{t-1} = n_1 = 2$ supersets to be reduced, each corresponding to the nodes {2.6560,4.2895} in stage 1, respectively. Therefore, w spans the range [1,2], and the supersets to be reduced are I_1^2 and I_2^2, respectively. I_1^2 corresponds to the first node, 2.6560, of the previous stage $t = 1$ (yellow points in Fig. 7(c)); therefore, I_1^2 comprises the values from that column of X pertaining to stage $t = 2$, whose row numbers are the profile numbers corresponding to the points in the first cluster C_1^1, i.e., $X_{[2,4,6,8,10],11}$. It is reduced to the optimal subset containing the points {3.2895,1.1758} (Fig. 7(d)). Similarly, I_2^2 corresponds to the second node, 4.2895, which is the cluster center of C_2^1), and contains the points $X_{[1,3,5,7,9],11}$ (Fig. 7(e)). It is reduced to the optimal subset {2.9582,5.6080}. The total number of leaf nodes at the final stage determines the number of profiles in the generated scenario tree (Fig. 7(f)); in this case, the generated scenario tree has 4 leaf nodes in the final stage $t = 2$.

5.2. Process distance for quality evaluation

Consider two probability distributions, $p \in P_1$ and $p' \in P_2$. Let the support sets for the distributions be Σ_1 and Σ_2, both defined on vector space Ξ. Then, the Kantorovich Distance (KD) between the distributions is the optimal objective function value of the optimization problem presented in Model (4). This does not take into account the amount of information being revealed at each stage, with respect to filtration.

The Kantorovich Distance (KD) can be extended to stochastic processes. Consider a stochastic process in finite time, ξ_t, where $t = 0, 1, 2..., T$. The information available at time t is denoted by $v_t = (\eta_0, \eta_1, \dots, \eta_t)$. Here, ξ_t can actually be denoted as a function of v_t. The sigma algebra generated by v_t is denoted by F_t. The sequence of increasing sigma algebras is known as a filtration, $F = (F_t)_{t=0}^T$.

The multistage distance, or the Process Distance (*PD*) is a more suitable metric to judge the similarity of two filtered probability measures, $p \in P_1$ and $p' \in P_2$, and is the optimal objective function value of the optimization problem (Pflug and Pichler, 2012; Beltrán et al., 2017),

$$PD = \min_{\pi_{i,j}} \sum_{i \in N_T, j \in N'_T} c_{i,j} \pi_{i,j} \tag{15a}$$

$$\text{s.t.} \quad \sum_{j \in N'_T : f \to j} \pi_{i,j} = \frac{p_i}{p_e} \times \sum_{i' \in N_T : e \to i'} \sum_{j' \in N'_T : f \to j'} \pi_{i',j'} \quad \forall p_i \in P_1, (e \to i, f) \tag{15b}$$

$$\sum_{i \in N_T : e \to i} \pi_{i,j} = \frac{p'_j}{p'_f} \times \sum_{i' \in N_T : e \to i'} \sum_{j' \in N'_T : f \to j'} \pi_{i',j'} \quad \forall p'_j \in P_2, (f \to j, e) \tag{15c}$$

$$\sum_{i,j} \pi_{i,j} = 1 \tag{15d}$$

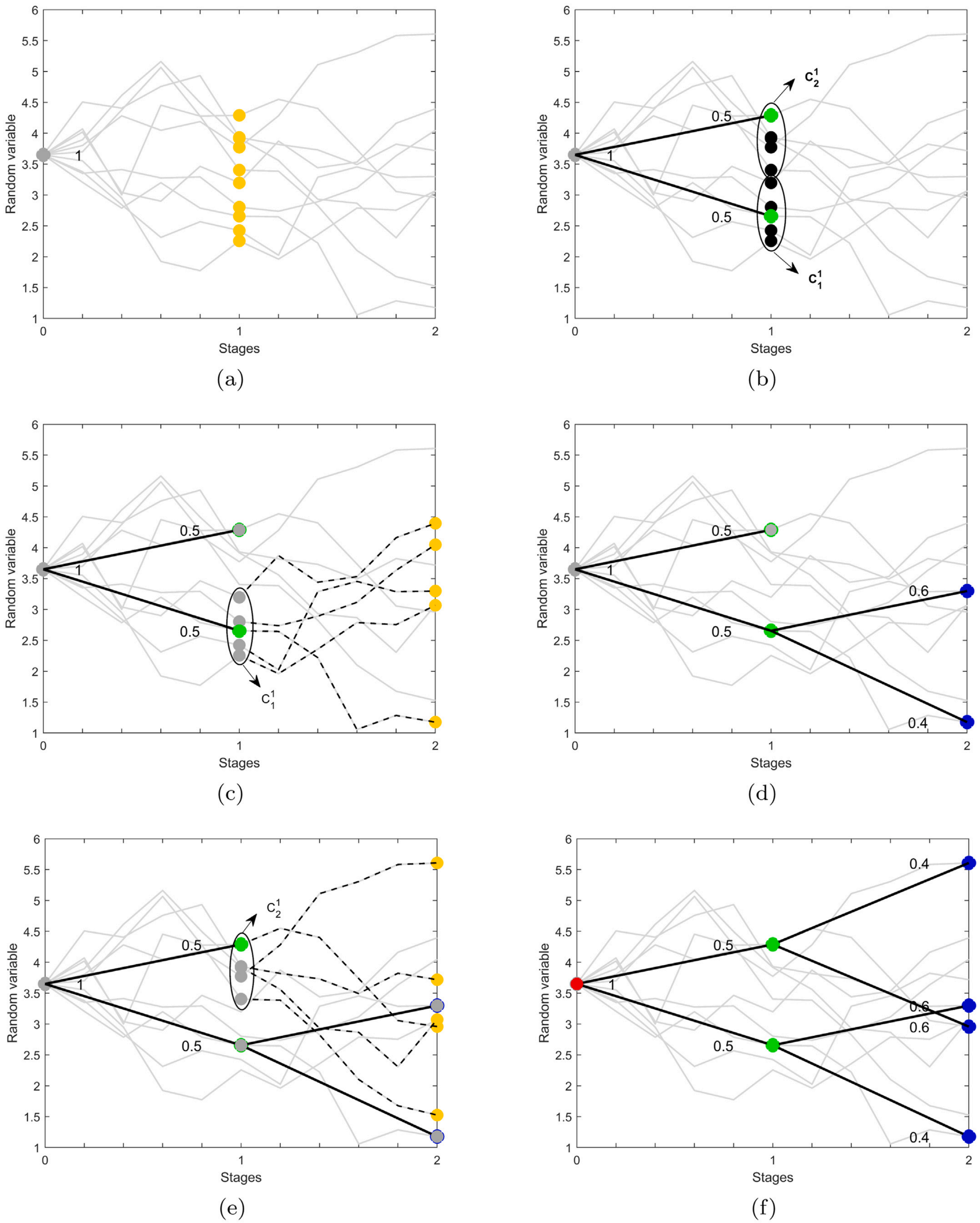


Fig. 7. Multistage scenario tree generation.

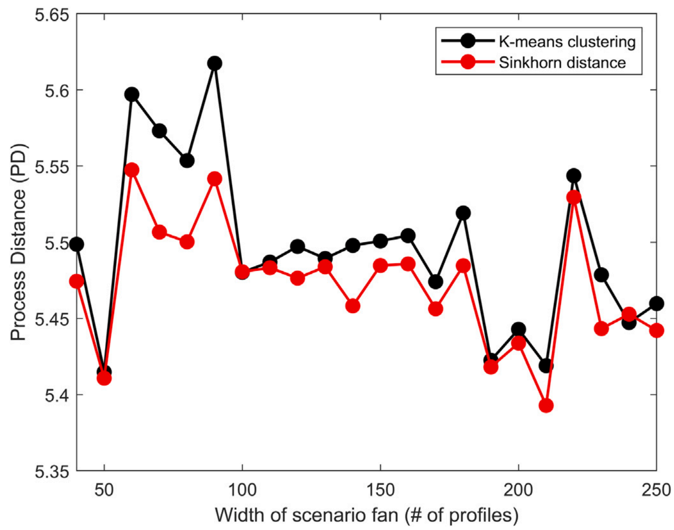


Fig. 8. A comparison of scenario tree generation approaches using process/multistage distance (PD) for a 5-(time) stage scenario tree.

$$\pi_{i,j} \geq 0 \qquad \forall i,j \tag{15e}$$

In Model (15), N_T and N'_T denote the set of all nodes at a given stage in the two scenario trees under consideration. The notations $f \to j$ and $e \to i$ denote that intermediate nodes f and e are predecessors of leaf nodes j and i. p_i, p'_j, p_e and p'_f are the probabilities of reaching nodes i, j, e, and f, respectively. The constraints (Eqs. (15b) and (15c)) impose the probability transfer between nodes of the scenario trees under comparison, at their corresponding time stages, subject to the satisfaction of conditional probability, by taking into account the sigma algebras involved.

The optimization problem in Model (15) accounts for probability mass transfer between corresponding time stages. This LP problem may become rather large, based on the number of stages, and the number of profiles in the original scenario fan, as well as the number of profiles in the reduced tree. It is solved in a decomposed manner, moving from one stage to the other.

Computational study

Fig. 8 depicts the performance of the Sinkhorn distance-based scenario reduction in contrast with the K-means clustering-based reduction (Xu et al., 2012) available in literature, using the process distance (PD) metric. In their work (Xu et al., 2012), the authors constructed a multistage scenario tree from a scenario fan by performing K-means clustering of the supersets at every stage t in order to obtain the reduced nodes. The reduced subset at every iteration is updated as the mean of the points in the K-means clusters. In the comparison between the Sinkhorn distance-based approach and the K-means based approach, the scenario fan width was varied between 40 and 250 profiles, and a 4-stage scenario tree was generated in every test case, containing 32 profiles (with $B_t = 2, t = 0, 1, \ldots, 4$). In order to generate these profiles, for every fan width, 100 test scenario fans were generated, each of which was converted into multistage scenario trees using the two methods, and their average approximation performance was plotted in the figure. It is observed that, overall, the proposed Sinkhorn distance-based scenario reduction algorithm gives better performance that the K-means clustering-based algorithm.

6. Case studies

In this section, we present two case studies to demonstrate the application of the proposed optimal scenario reduction and scenario tree generation approaches. The first case study is a two-stage stochastic programming-based planning problem where the scenario reduction approach from Section 4 is applied. The second case study is a multistage stochastic programming-based chemical plant design problem where the scenario tree generation method from Section 5 is applied.

6.1. The farmer's problem

The farmer's problem (Birge and Louveaux, 2011) presents a benchmark case study for stochastic programming in literature; in this work, we illustrate the performance of the proposed Sinkhorn distance-based scenario reduction algorithm developed on this study. The problem deals with the allotment of 500 acres of land as farming area to be distributed between a choice of three crop species – wheat, corn, and sugar beet – under uncertainty in crop yield (y_1, y_2, y_3). To this end, x_1, x_2, and x_3 are defined as the amounts of land (in acres) allotted to wheat, corn, and sugar beet, respectively. The planting costs per acre are given as \$150, \$230, and \$260, respectively. Furthermore, constraints are placed on the problem as follows,

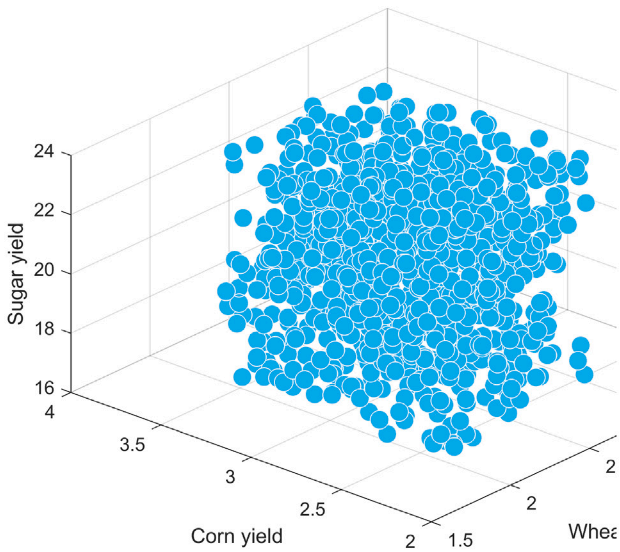
- Since wheat and corn are also used as cattle feed, the minimum required yield of wheat and corn are 200 tons and 240 tons, respectively.
- Wheat may be purchased at a price of \$238 per ton, while corn may be purchased for \$210 per ton. In this problem, b_1 and b_2 are defined as the amount of wheat and corn purchased (in tons), respectively.
- Wheat and corn may be sold at \$170 and \$150 per ton, respectively. Sugar beet may be sold at a price of \$36 per ton for amounts under 6000 tons, and at a lower price of \$10 per ton thereafter. In this problem, s_1 and s_2 are defined as the amount of wheat and corn sold (in tons), respectively. The amount of sugar beet sold at the higher price is defined as s_3, whereas that sold at the lower price is defined as s_4.

The optimization model for the farmer's problem may be written as:

$$\begin{aligned}
\min\ & 150x_1 + 230x_2 + 260x_3 + 238b_1 - 170s_1 + 210b_2 - 150s_2 \\
& - 36s_3 - 10s_4 && (16a)\\
\text{s.t. } & x_1 + x_2 + x_3 \leq 500 && (16b)\\
& y_1x_1 + b_1 - s_1 \geq 200 && (16c)\\
& y_2x_2 + b_2 - s_2 \geq 240 && (16d)\\
& s_3 + s_4 \leq y_3x_3 && (16e)\\
& s_3 \leq 6000 && (16f)\\
& x_1, x_2, x_3, b_1, b_2, s_1, s_2, s_3, s_4 \geq 0 && (16g)
\end{aligned}$$

In the scenario-based stochastic programming formulation of the problem, a finite set (K) of scenarios of crop yield $(y_{1,k}, y_{2,k}, y_{3,k} \quad \forall k \in K)$ are considered. These scenarios are assumed to be based around the average yield of 2.5, 3, and 20 for wheat, corn, and sugar beet, respectively. To this end, 1000 representative scenarios are generated, as shown in Fig. 9(a). The scenario-based stochastic programming formulation of the farmer's problem is given in Model (17) as,

$$\begin{aligned}
\min\ & 150x_1 + 230x_2 + 260x_3 + \sum_{k\in K} p_k \Big(238b_{1,k} \\
& - 170s_{1,k} + 210b_{2,k} - 150s_{2,k} - 36s_{3,k} - 10s_{4,k} \Big) && (17a)\\
\text{s.t. } & x_1 + x_2 + x_3 \leq 500 && (17b)\\
& y_{1,k}x_1 + b_{1,k} - s_{1,k} \geq 200, \quad \forall k \in K && (17c)\\
& y_{2,k}x_2 + b_{2,k} - s_{2,k} \geq 240, \quad \forall k \in K && (17d)\\
& s_{3,k} + s_{4,k} \leq y_{3,k}x_3, \quad \forall k \in K && (17e)\\
& s_{3,k} \leq 6000, \quad \forall k \in K && (17f)
\end{aligned}$$



(a)

(b)

Fig. 9. Scenario representation of 3-dimensional uncertain yield (a) 1000 scenario superset, (b) 50 scenario subset from the proposed Sinkhorn distance-based scenario reduction method (Algorithm 2)

$$x_1, x_2, x_3 \geq 0 \tag{17g}$$

$$b_{1,k}, b_{2,k}, s_{1,k}, s_{2,k}, s_{3,k}, s_{4,k} \geq 0, \quad \forall k \in K \tag{17h}$$

Here, p_k refers to the probability of occurrence of scenario k. The probability values (p_k) are computed by the Sinkhorn distance-based scenario reduction algorithm as the probability mass allocated to each element in the set of reduced scenarios from the superset of elements.

The stochastic optimization model using the scenarios generated from the proposed Sinkhorn distance-based method was solved using the PySP (Watson et al., 2012) module in Python, for 30 different runs, for the same original superset of 1000 scenarios. The solutions from each of these stochastic optimization runs was extracted, and the optimal objective values were calculated.

The quality of the generated scenario set is judged by two tests: in-sample stability and out-of-sample stability (Kaut and Wallace, 2003). It is important to note here, that how good a scenario generation method is does not depend on how well-approximated the scenario tree is with respect to the original continuous scenario distribution; rather, it depends on how good the solution of the model using the scenario tree is, with respect to the solution using the original model.

The in-sample stability test for a scenario generation method requires that, "for a number of scenario trees generated by the chosen scenario generation method, whichever scenario tree is chosen, the optimal objective value reported across these multiple scenario trees is approximately the same". In-sample stability of a scenario generation method ensures that each time the method is used to generate a set of scenarios, it always performs to give similar optimal objective values. In-sample stability does not give any insight into how good the stochastic approximation of the model itself is, with respect to the true model; rather, it is only concerned with achieving a similar level of good each time it is used for scenario generation for a particular problem statement.

In contrast to the in-sample stability test, which compares the performance of the scenario generation method across different iterations of the method itself, the out-of-sample stability test compares the performance of the scenario generation method to the performance of the actual model itself. The out-of-sample stability test for a scenario generation method requires that, "for a number of scenario sets (trees) generated by the chosen generation method, whichever scenario tree is chosen, the optimal solution reported by each of these scenario sets is approximately equal to the true solution itself, and by extension, the optimal objective value reported by each of these multiple scenario sets is approximately equal to the true optimal objective value". In contrast to in-sample stability, out-of-sample stability does give insight into how good the stochastic approximation of the model itself is, with respect to the true model — it gives insight into how good the scenario generation method is for a particular problem statement.

The in-sample, and out-of sample stability results for this case study are depicted in Figs. 10(a) and 10(b), respectively. Both stability tests were conducted for a number of reduced scenario subset sizes of 50, 100, 150, 200, 250, and 500 scenarios, for the same reference scenario superset of 1000 scenarios. From Fig. 10(a), it is observed that as the number of representative scenarios in the reduced subset increases, the variance in the optimal objective value decreases overall. From Fig. 10(b), it is observed that when the true problem is solved for first stage decisions fixed from the solutions of the approximated problems, the optimal objective on average reaches close to the true optimal objective value of 1.113×10^5.

6.2. Chemical plant design problem

In this section, we illustrate the use of optimal scenario trees generated using Algorithm 3 in the multiperiod design and operation of a chemical plant under demand uncertainty, adapted from the work of Subrahmanyam et al. (1994).

The design superproblem in this case study is defined over the set of all tasks to be performed, denoted by $i \in I$, the set of all plant types that can be considered for construction denoted by $j \in J$, the set of all resources in play is denoted by $s \in S$, and the set of all time nodes is denoted by $t \in T$. Furthermore, I_j^{tasks} is the set of tasks that can be performed by each plant type, while I_i^{equip} is the set of plant types that each task can be performed on.

The model contains a number of decision variables defined over the aforementioned sets. $y_{i,j,t}$ denotes the number of times task i is performed on plant type j during the time period t. Each time period is H_t days long, and each task takes $p_{i,j}$ days to complete. The number of plants of type j that come online in time period t is denoted by $n_{j,t}$, while the total number of plants of type j that are active in the time

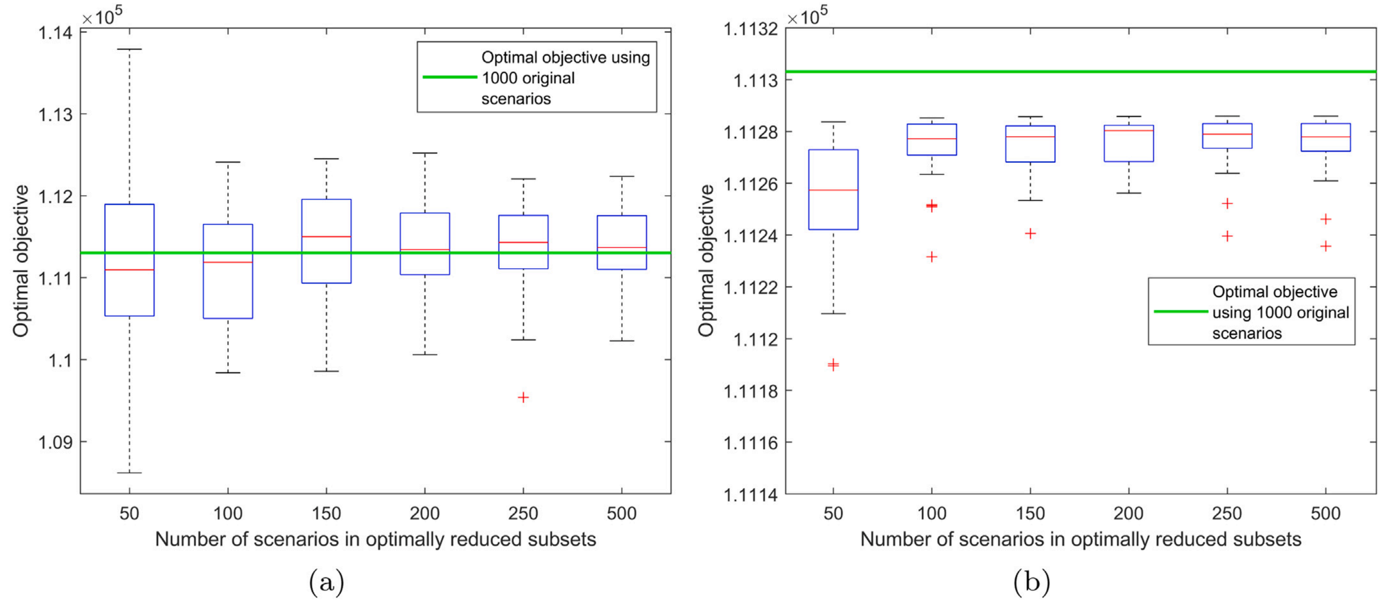


Fig. 10. Stability test results for the 2-stage farmer's problem: (a) In-sample stability, (b) Out-of-sample stability.

period t is denoted by $N_{j,t}$. The cost incurred in performing tasks is denoted by $c^o_{i,j,t}$, and the total operational budget for each time period is given by C^o_t. The amount of each task that is performed on each plant type is arbitrarily measured in reaction units, and is denoted by $B_{i,j,t}$. The capacity of each plant to perform a task is given by $m_{i,j}$. The amount of resource s that is available in inventory at the end of time period t is denoted by $A_{s,t}$, and is estimated by a material balance, where the stoichiometric ratio is given by $f_{s,i}$. The amount of resource s that is sold in time period t is given by $q_{s,t}$ and the amount purchased by $z_{s,t}$. It is to be noted that only certain resources ($s = 4, 7$) are sold, and only some resources ($s = 1, 2$) are purchased. The limit on the amount of resources that can be stored in inventory at the end of any time period is given by $A^{\max}_{s,t}$. The maximum purchase limit on resources is given by $Z_{s,t}$.

Additionally, the amount of resources sold, $q_{s,t}$, is classified into two types, the amount sold below and equal to the demand, denoted by $q^0_{s,t}$, and the amount sold exceeding demand, given by $q^+_{s,t}$. Only the amount of resources sold below and equal to the demand contributes to the total profit. In this problem, the demand for resources is stochastic in nature, and takes the value $Q_{s,k,t}$, where $k \in K$ represents the set of discrete uncertain scenarios. The parameters used in the problem are given in Table 3. The demand for resources $Q_{4,k,t}$ and $Q_{7,k,t}$ across k scenarios is uniformly distributed between with an average of $[150, 200]$, respectively, and deviation of 7, across 1000 scenarios, which is further reduced optimally to smaller subset sizes. The complete stochastic optimization problem is formulated as Model (18).

$$\min \sum_{t=1}^{T} \mathbb{E}_k \left[\sum_{s=1}^{S} (v^{\text{sold}}_{s,t} q^0_{s,k,t} - v^{\text{buy}}_{s,t} z_{s,k,t}) - \sum_{j=1}^{J} \left(n_{j,k,t} C_{j,t} + \sum_{i=1}^{I} c^o_{i,j,t} y_{i,j,k,t} \right) \right] \tag{18a}$$

$$\text{s.t.} \quad \sum_{i \in I^{\text{tasks}}_j} p_{i,j} y_{i,j,k,t} \le H_t N_{j,t} \qquad \forall j \in J, t \in T, k \in K \tag{18b}$$

$$N_{j,t} = \sum_{\tau=1}^{t} n_{j,k,\tau} \qquad \forall j \in J, t \in T, k \in K \tag{18c}$$

$$A_{s,k,t} = A_{s,k,t-1} + \sum_i \sum_{j \in I^{\text{equip}}_i} f_{s,i} B_{i,j,k,t} - q_{s,k,t} + z_{s,k,t} \qquad \forall s \in S, t \in T, k \in K \tag{18d}$$

$$A_{s,k,t} \le A^{\max}_{s,t} \qquad \forall s \in S, t \in T, k \in K \tag{18e}$$

$$B_{i,j,k,t} \le m_{i,j} y_{i,j,k,t} \qquad \forall i \in I, j \in J, t \in T, k \in K \tag{18f}$$

$$q_{s,k,t} = q^0_{s,k,t} + q^+_{s,k,t} \qquad \forall s \in S, t \in T, k \in K \tag{18g}$$

$$q_{s,k,t} \le Q_{s,k,t} \qquad \forall s \in S, t \in T, k \in K \tag{18h}$$

$$z_{s,k,t} \le Z_{s,t} \qquad \forall s \in S, t \in T, k \in K \tag{18i}$$

$$A_{s,k,t}, z_{s,k,t}, q^0_{s,k,t}, q^+_{s,k,t}, B_{i,j,k,t} \in \mathbb{R}^+ \qquad \forall i \in I, j \in J, s \in S, t \in T, k \in K \tag{18j}$$

$$n_{j,k,t}, y_{i,j,k,t} \in \mathbb{Z}^+ \qquad \forall i \in I, j \in J, t \in T, k \in K \tag{18k}$$

$$N_{j,t} \in \mathbb{Z}^+ \qquad \forall j \in J, t \in T \tag{18l}$$

As in the case of the farmer's problem case study, we solved this multistage stochastic programming problem using the PySP (Watson et al., 2012) module in Python. The PySP module generates a stochastic formulation of the optimization model, given the scenario structure, containing the following correspondence information: (1) variable-stage, (2) parent node-leaf node, (3) node-stage, (4) scenario-leaf node at final stage.

In this case study, we assume that the true distribution of the resource demands $Q_{s,t}$ may be represented by a set of 1000 scenarios. In order to evaluate the performance of the proposed algorithm for scenario tree generation, a number of scenario trees containing 50, 100, 150, 200, 250, and 500 profiles, respectively, were generated, and the scenario-based stochastic programming problem was solved. Fig. 11 shows the in-sample stability results for the chemical plant case study. From the figure, we observe that as the number of scenarios in the reduced scenario tree increases from 50 to 500, the objective function value gets closer to the true objective function value of 2.6667×10^4. Fig. 12 shows the evolution of process distance with tree size. As the number of approximating scenarios increases, the process distance decreases and gets closer to the PD value of a 1000 scenario tree.

7. Conclusion

In this work, we propose a method to reduce a large set of scenarios to a smaller subset of chosen size through entropy-regularized optimal

Table 3
List of parameters for the chemical plant design problem.

Parameter		Time period				Remarks
		$t = 1$	$t = 2$	$t = 3$		
$A^{max}_{s,t}$	$s = 1$					
	$s = 2$					
	$s = 3$					
	$s = 4$		400			
	$s = 5$					
	$s = 6$					
	$s = 7$					
H_t		80	80	80		
$C_{j,t}$	$j = 1$					
	$j = 2$					
	$j = 3$					
$Z_{s,t}$	$s = 1$					
	$s = 2$					
$v^{sold}_{4,t}$		51	50	49		$v^{sold}_{s,t} = 0, \forall s = 1, 2, 3, 5, 6$
$v^{sold}_{7,t}$		70	71	68		
$v^{buy}_{1,t}$		23	24	25		$v^{buy}_{s,t} = 10^{20}, \forall s = 3, 4, 5, 6, 7$
$v^{buy}_{2,t}$		25	26	27		
Parameter		Plants				
		$j = 1$	$j = 2$	$j = 3$		
$p_{i,j}$	$i = 1$					
	$i = 2$		4			
	$i = 3$					
	$i = 4$					
$m_{i,j}$	$i = 1$					
	$i = 2$	100	200	150		
	$i = 3$					
	$i = 4$					
I^{tasks}_j		1,4	1,4	2,3		
Parameter		Tasks				
		$i = 1$	$i = 2$	$i = 3$	$i = 4$	
I^{equip}_i		1,2	3	3	1,2	
$f_{s,i}$	$s = 1$	−1	0	0	0	
	$s = 2$	−1	0	−1	0	
	$s = 3$	1	−1	0	0	
	$s = 4$	0	1	0	0	
	$s = 5$	0	1	−1	0	
	$s = 6$	0	0	1	−1	
	$s = 7$	0	0	0	1	

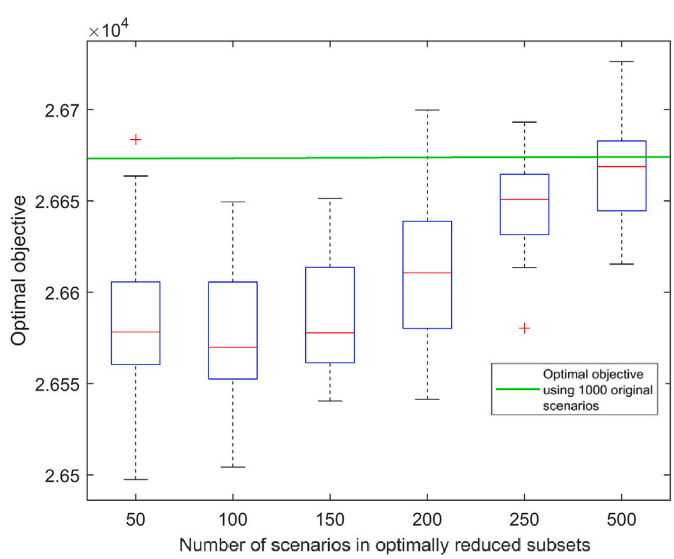


Fig. 11. In-sample stability results for multi-period stochastic design of a chemical plant.

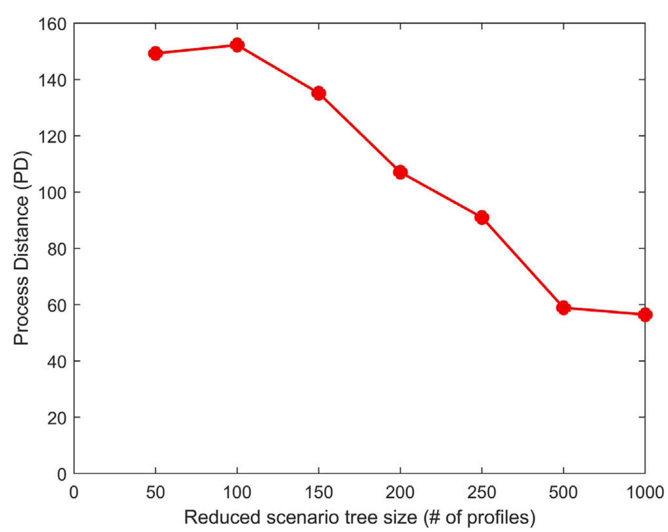


Fig. 12. Process distance evolution with reduced scenario set size for multi-period stochastic design of a chemical plant.

transport. We leveraged the Sinkhorn–Knopp algorithm to replace the linear programming formulation of the discrete optimal transport problem with an iterative computational procedure, to obtain the Sinkhorn distance as a quantitative measure of similarity between the probability distributions of the superset and the optimal subset. Using this measure, we developed an algorithm for optimal scenario reduction, and extended its use to scenario tree generation for multistage stochastic programming. We illustrated the proposed scenario reduction and scenario tree generation algorithms on two case studies, and assessed their stability. We found that replacing the Kantorovich Distance with the Sinkhorn distance in scenario reduction resulted in a significant decrease in computational time. Furthermore, in stochastic programming applications, the solutions obtained from using Sinkhorn distance-based scenario reduction converged to the solutions of the problems solved using the original scenario set with good accuracy. The proposed algorithms for scenario reduction and scenario tree generation are effective for scenario-based stochastic programming.

CRediT authorship contribution statement

Sanjula Kammammettu: Methodology, Software, Writing – original draft. **Zukui Li:** Conceptualization, Supervision, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Zukui Li reports financial support was provided by Natural Sciences and Engineering Resource Council (NSERC) of Canada.

Data availability

Data will be made available on request.

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